# ECE343 Spring 2025

## Objective 3.56: Smith Charts & Impedance Matching



| Lesson | Topics | Reading | Notes |
|---|---|---|---|
| 1 | Course Overview & Introduction | Ch.1 | LO 1.1-1.2 |
| **Chapter 2: Electric and Magnetic Fields** | | | |
| 2 | Fields, Parameters, Field Quantities | Ch. 2 | LO 2.1-2.2 |
| **Chapter 3: Transmission Lines** | | | |
| 3 | Lumped Element Model | 3.1-3.4 | LO 3.1 |
| 4 | Telegrapher's Equations | 3.5-3.6 | LO 3.1 |
| 5 | Characteristic Impedance & Wave Propagation | 3.7-3.9 | LO 3.2 |
| 6 | Coaxial, Reflection Coefficient, VSWR | 3.10,3.12-3.14 | LO 3.2-3.3 |
| 7 | Terminated Transmission Lines | 3.15,3.17-3.19 | LO 3.4 |
| 8 | Impedance Matching | 3.21-3.23 | LO 3.4 |
| 9 | Impedance Matching | 3.21-3.23 | LO 3.4 |
| 10 | Smith Charts | Supplemental | LO 3.5 |
| 11 | Smith Charts | Supplemental | LO 3.5 |
| 12 | Smith Charts | Supplemental | LO 3.5 |
| 13 | Smith Charts | Supplemental | LO 3.5 |
| 14 | Transients | Supplemental | LO 3.6 |
| 15 | Reflection/Transmission Coefficient Lab | Lab Packet | LO 3.7 |
| **Chapter 4: Vector Analysis** | | | |
| 16 | Vector & Coordinate System Review | Ch. 4 | LO 4.1-4.4 **GR 1 due** |
| **Chapter 5: Electrostatics** | | | |
| 17 | Coulomb's Law | 5.1-5.3 | LO 5.1 |
| 18 | Surface and Volume Charge | 5.4 | LO 5.1 |
| 19 | Gauss' Law (Integral Form) | 5.5-5.6 | LO 5.2 |
| 20 | Gauss' Law (Differential Form) | 5.7-5.8 | LO 5.2 |
| 21 | Electric Potential | 5.9-5.10, 5.15 | LO 5.3 |

# Ch 3 Learning Objectives

3.1 I can describe the origin of the transmission line equation, understand the method of developing a solution, and interpret the solution.

3.2 I can explain the physical meaning of and calculate the following quantities: propagation constant, phase constant, attenuation constant, characteristic impedance, VSWR, reflection coefficient, and transmission coefficient.

3.3 I can use the transmission line equation to calculate the voltage/current at an arbitrary point on a specified transmission line.

3.4 I can use the transmission line equation to match a load to a source using a length of transmission line.

3.5 I can identify the various quantities represented on the Smith Chart.

3.6 I can use a Smith Chart to design a matching network for a given source and load, using series components, parallel components, shorted stubs, and open stubs.

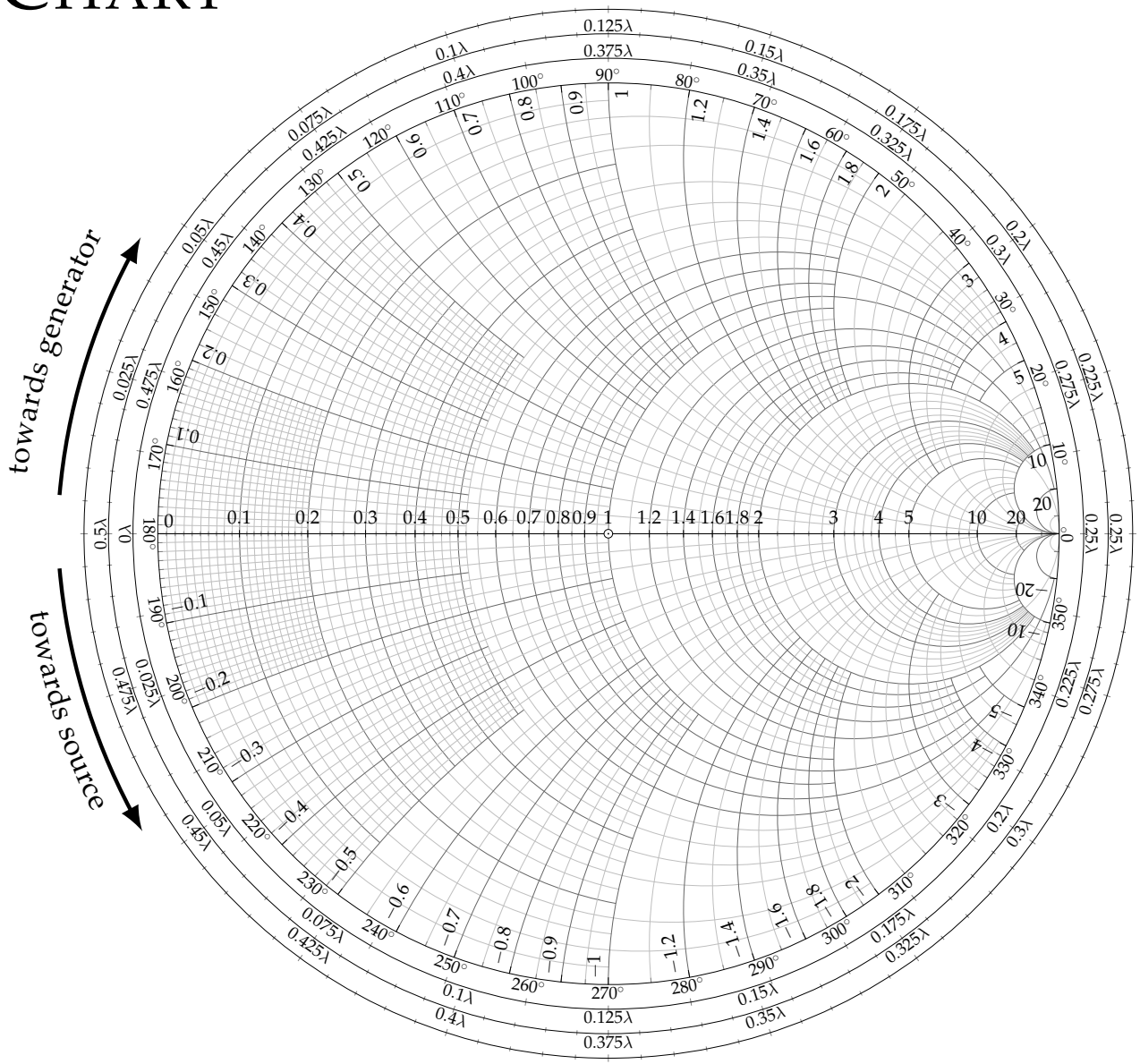
3.7 I can plot the transient response of a load connected to a source through a transmission line.

3.8 I can measure the reflection coefficient across a range of frequencies and interpret the measurement using a network analyzer.

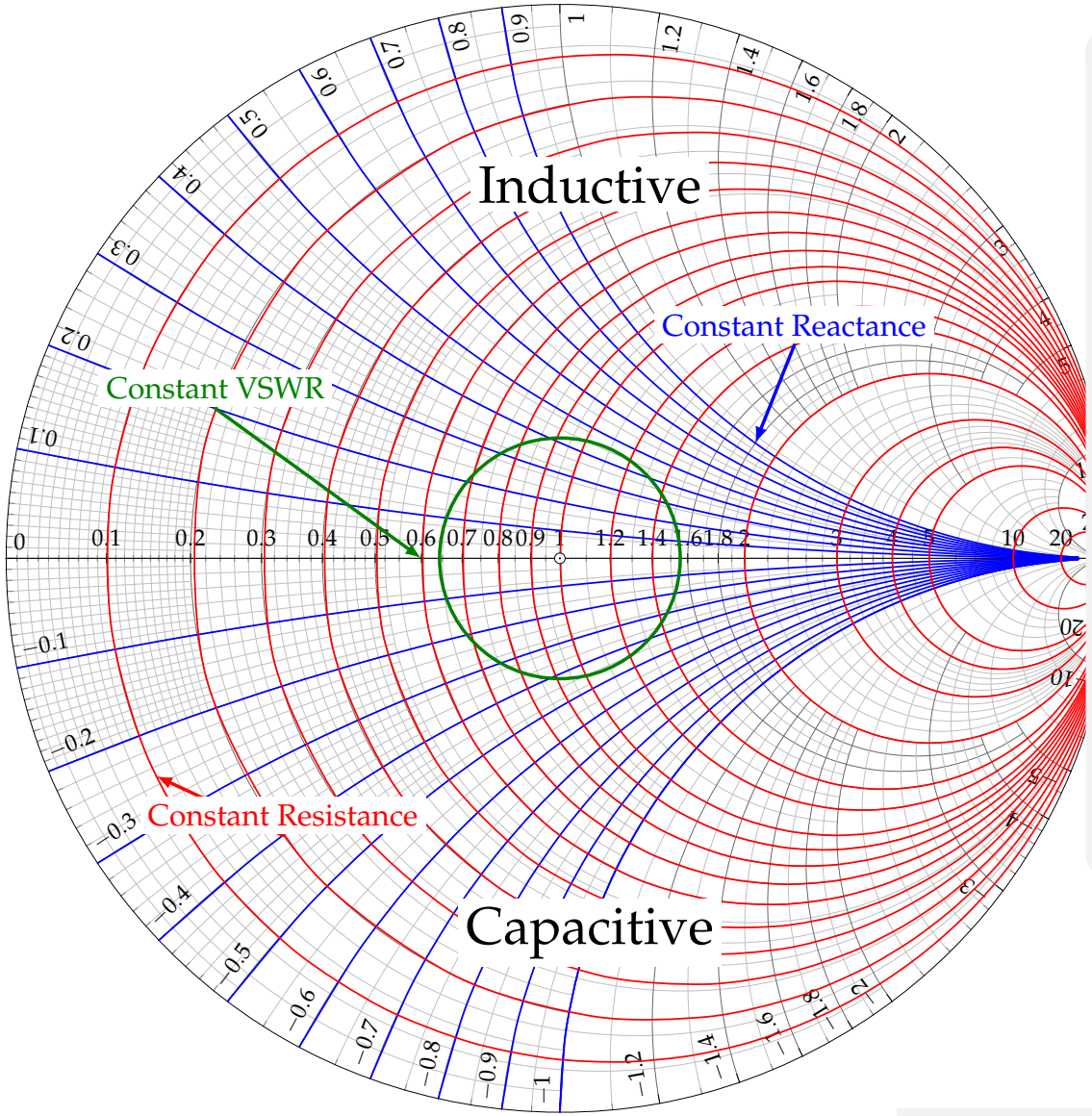
# Smith Charts: History

- Developed in 1939 by Phillip H. Smith
- Used to solve a variety of tx line and waveguide problems
- Measure/evaluate complex reflection/transmission on a tx line
- Measure/evaluate VSWR
- Measure/evalute impedance along a tx line
- Evaluate IO transfer functions

# Smith Chart



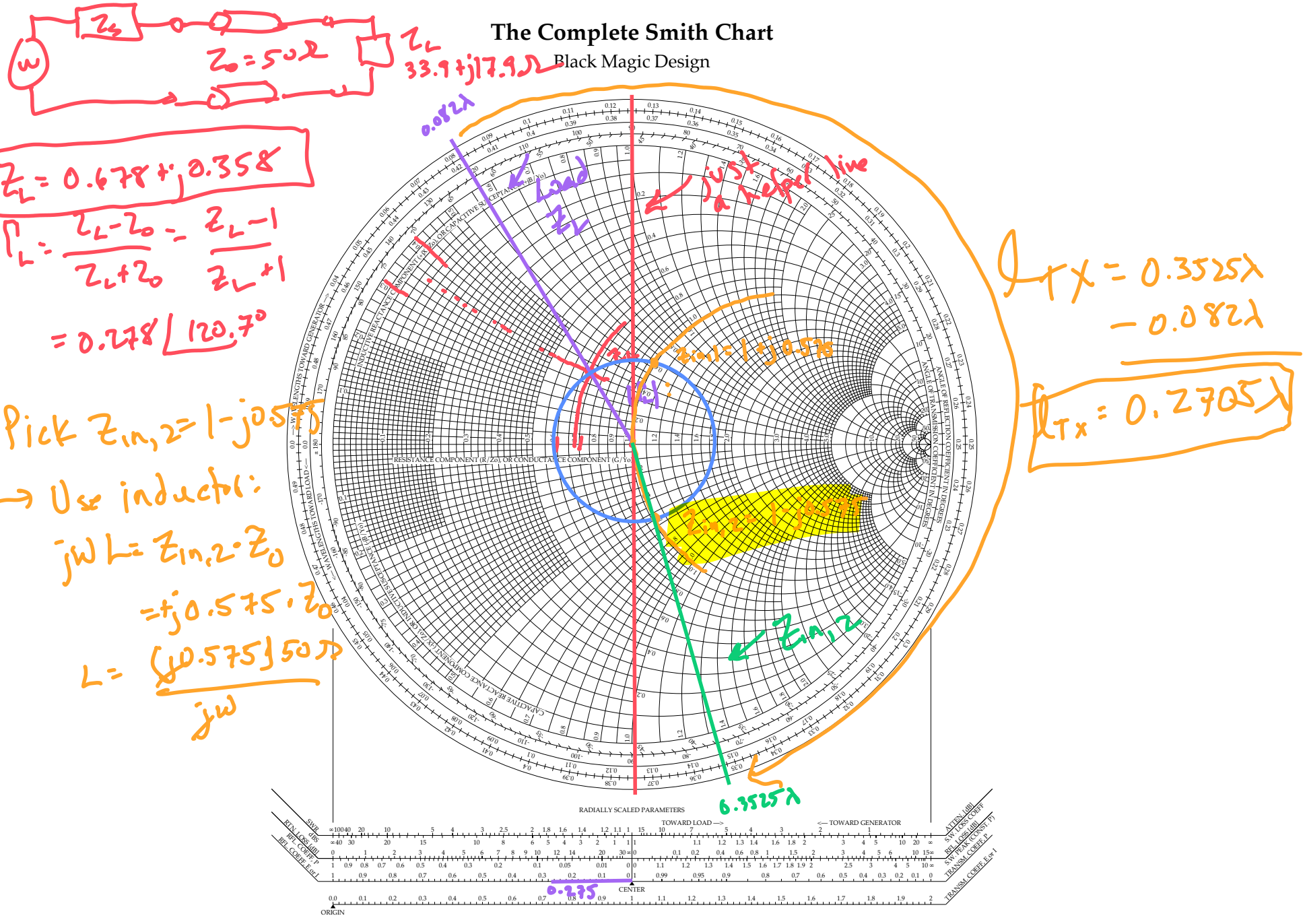
# Smith Charts: Measurements




- Plot complex impedances
- Use circles of constant R & X
- Measure $\Gamma_L$
- Magnitude and Phase
-

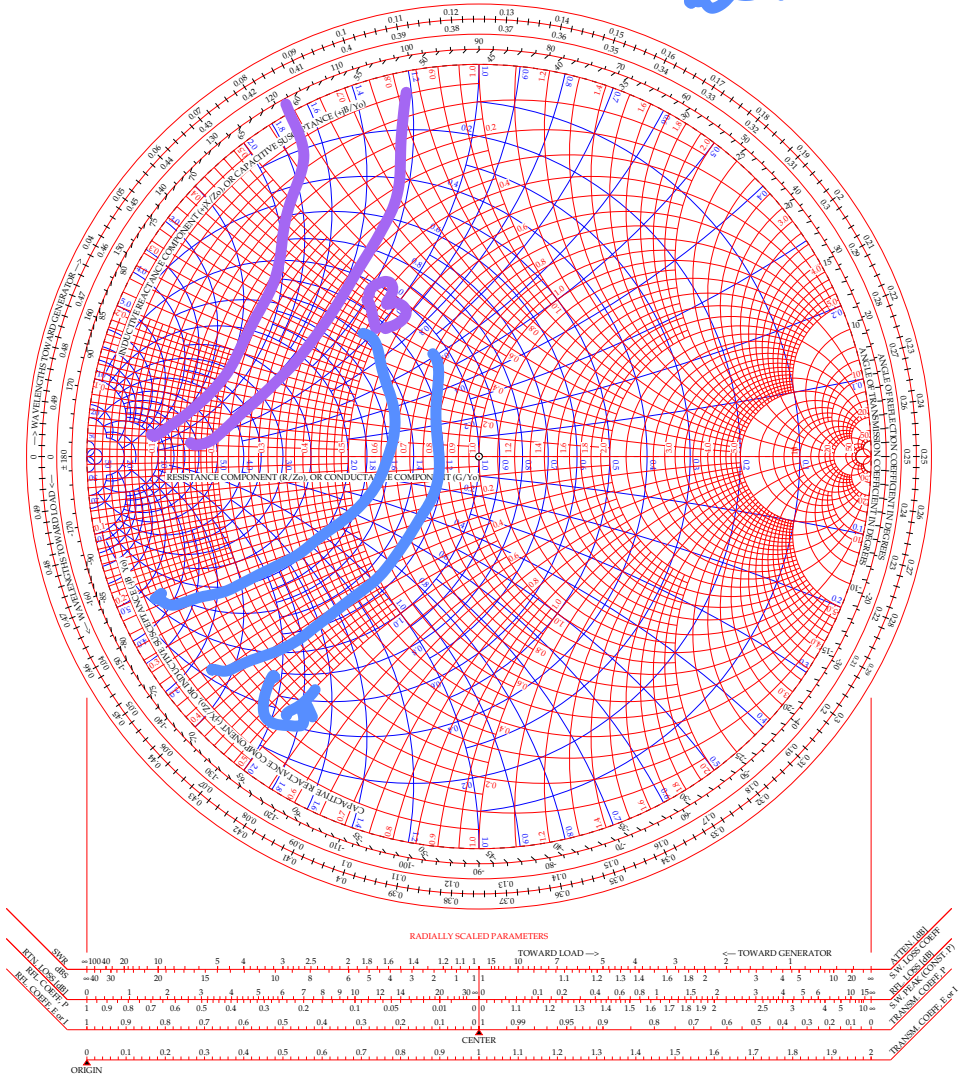
# The Complete Smith Chart

Black Magic Design

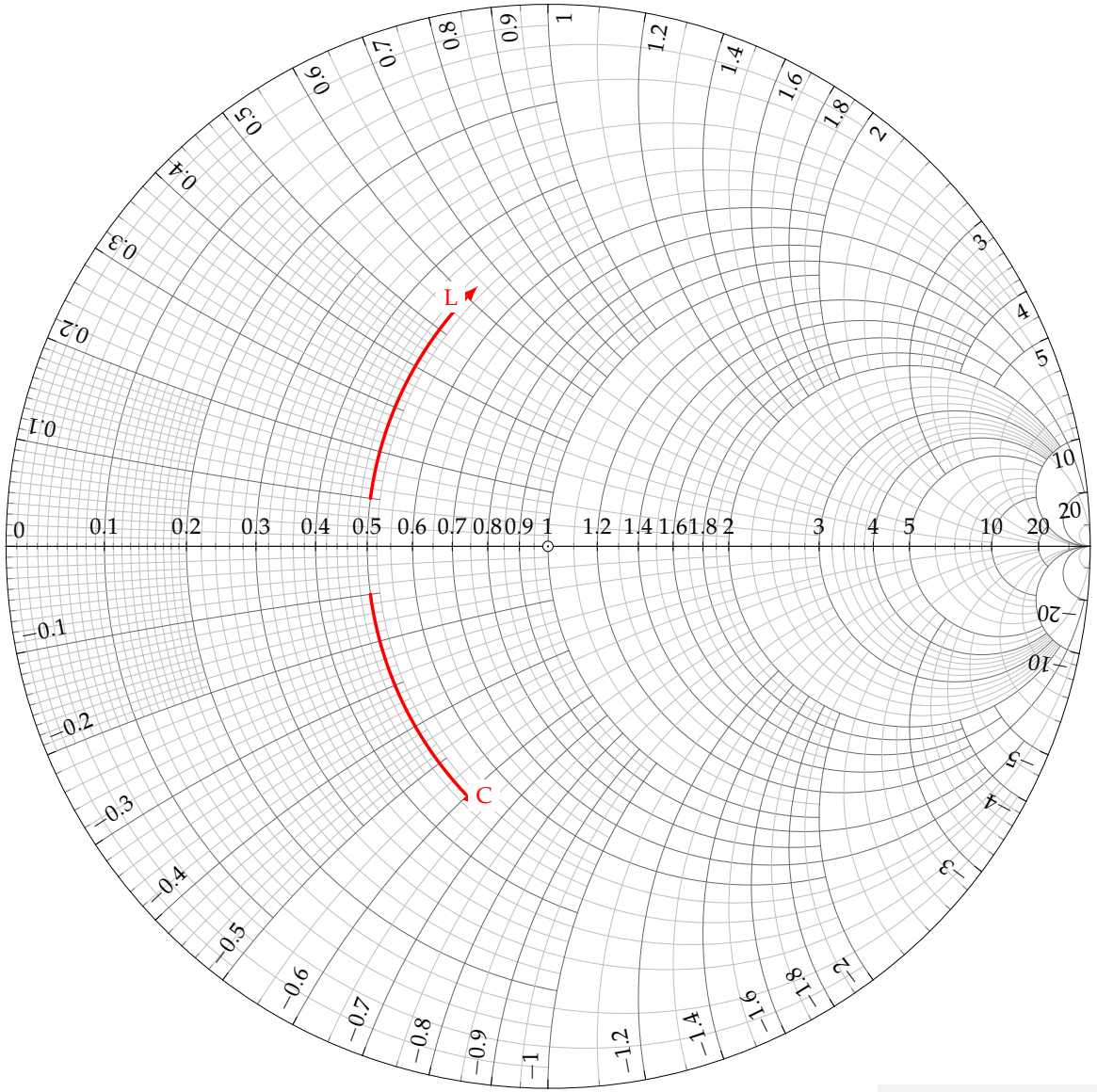


$Z = R + jX$

$Y = G + jB$


RESISTANCE COMPONENT (R/Zo), OR CONDUCTANCE COMPONENT (G/Yo)
REACTANCE COMPONENT (jX/Zo), OR SUSCEPTANCE COMPONENT (jB/Yo)
RADIALLY SCALED PARAMETERS
TOWARD LOAD —>
<— TOWARD GENERATOR
CENTER
ORIGIN

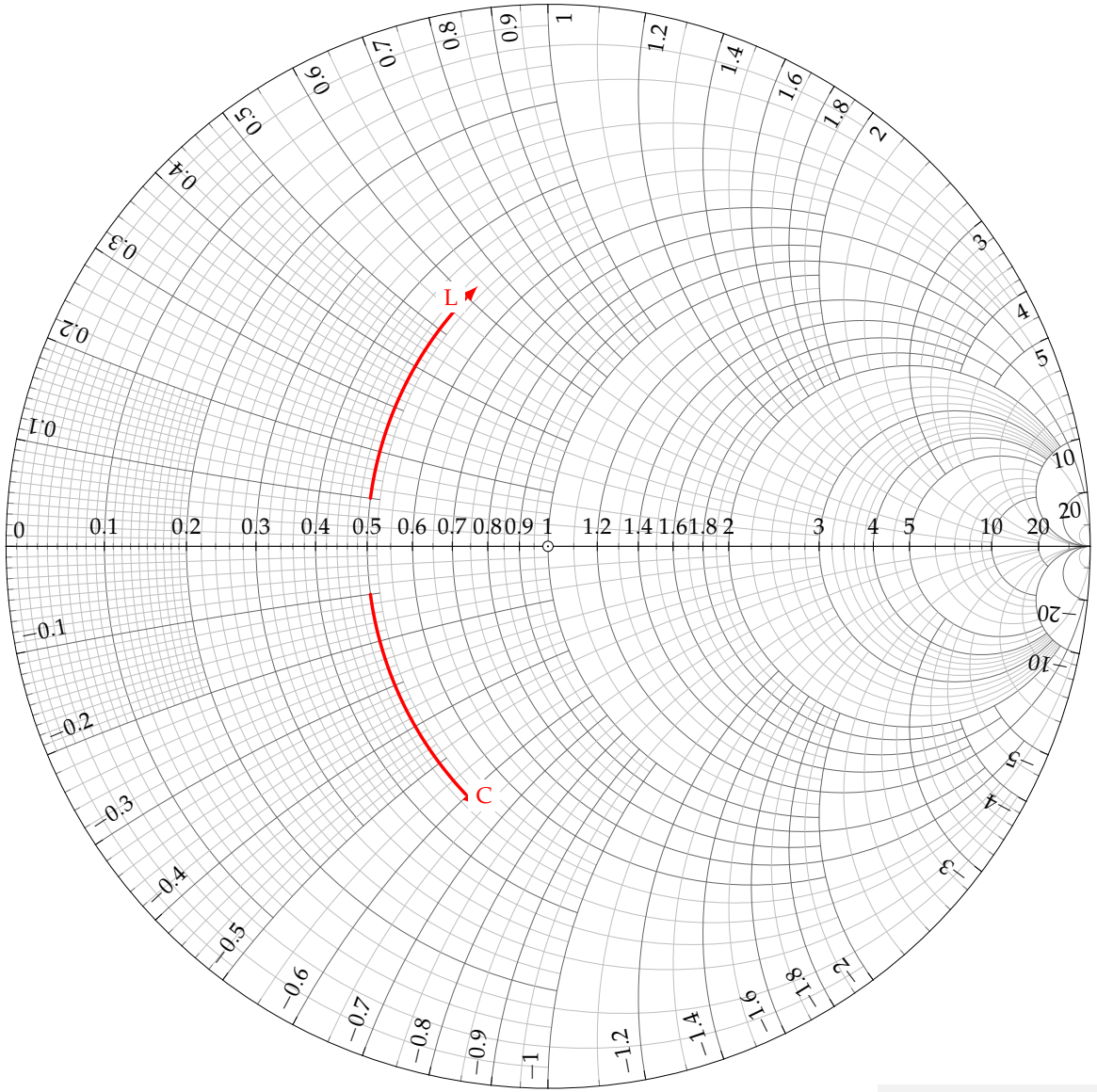
# Smith Charts: Components




## Series

- Series _____ move _____ along lines of constant resistance
- Series _____ move _____ along lines of constant resistance

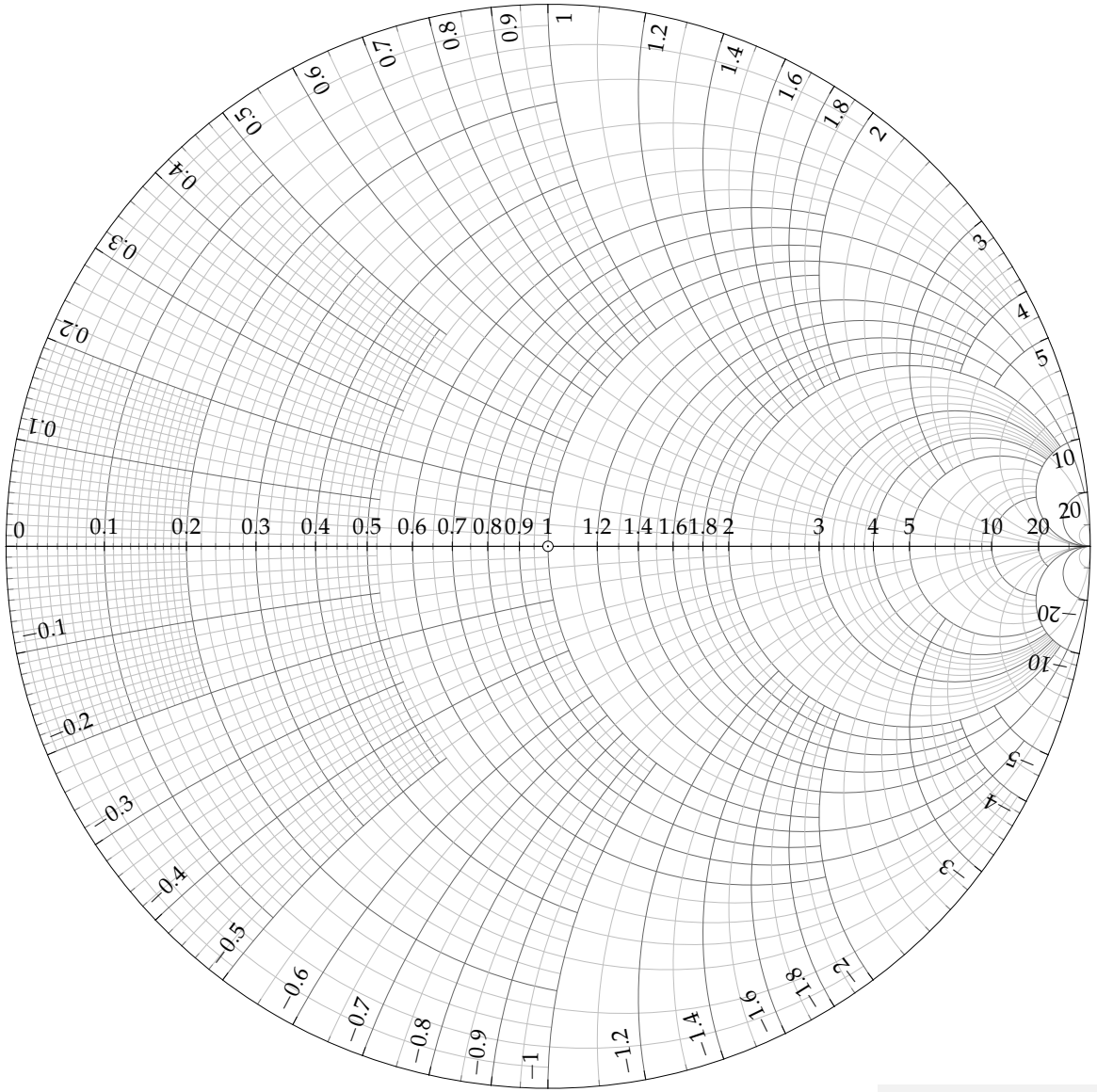
# Smith Charts: Components




## Parallel

- Parallel _____ move _____ along lines of constant conductance
- Parallel _____ move _____ along lines of constant conductance

# Smith Charts: Plotting Impedances

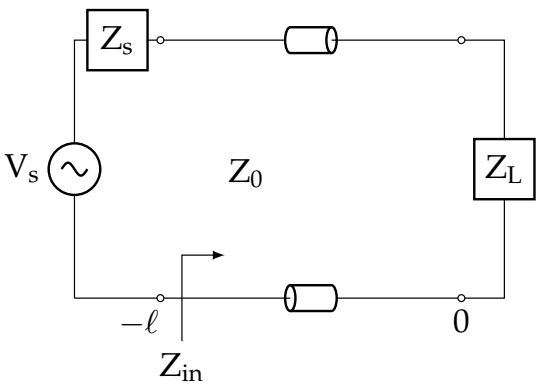



## Plot these

- $Z_L = 2 - j1.4$ $\longrightarrow$ (C or L?)
- $Z_L = 1.5 + j0.5$ $\longrightarrow$ (C or L?)
- $Z_L = 75 + j60\Omega$ with $Z_0 = 50\Omega$
- Short circuit
- Open circuit

# Impedance Matching: General Process

Match the combined load ($Z_L$ and TX line) to $Z_s$.




## Process

1. Normalize impedances to $Z_0$
2. Find reflection coefficient ($\Gamma_L$)
3. Holding the Mag. of $\Gamma_L$ constant, find the input impedance $z_{in} =$ _____

$$\Longrightarrow |\Gamma_L| = \frac{z_{in} - 1}{z_{in} + 1}$$

$\mathrm{Re}[z_{in}] = z_s$ only real

4. Determine capacitance or inductance from $X$.

$$\Longrightarrow Z_C = \frac{1}{j\omega C} \text{ or } Z_L = j\omega L$$

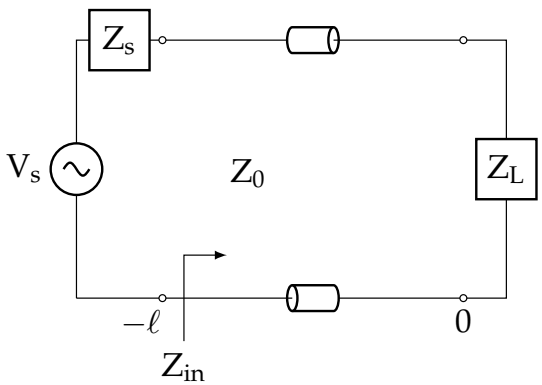
$z_{in} = z_s \pm jX$

5. Find the length of TX line to make:

$$\Longrightarrow z_{in} = 1 + jX = \frac{z_L + j\tan(\beta\ell)}{1 + jz_L\tan(\beta\ell)}$$

# Impedance Matching: Smith Chart Process

Match the combined load ($Z_L$ and TX line) to $Z_s$.




## Process

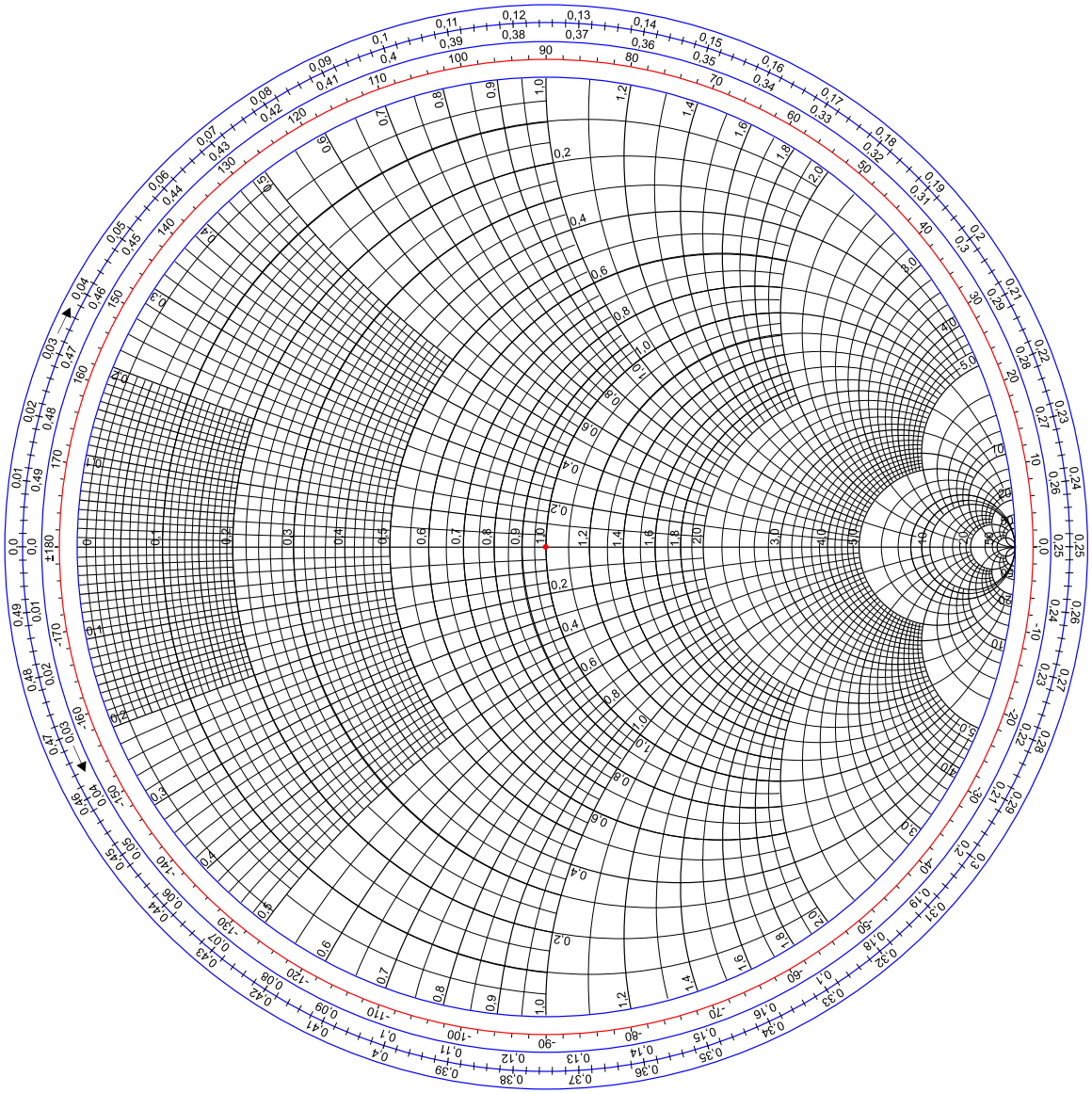
1. Normalize impedances to $Z_0$, plot $Z_L$
2. Draw circle of constant $|\Gamma_L|$ (life)
3. Rotate towards source on circle until intersection with circle of constant R → $Z_s$
4. Read $z_{in,1}$ or $z_{in,2}$
5. Determine capacitance or inductance from $X$.

$$\Longrightarrow Z_C = \frac{1}{j\omega C} \text{ or } Z_L = j\omega L$$

Unitize!

6. Measure TX line length ($\ell$) from outer scale

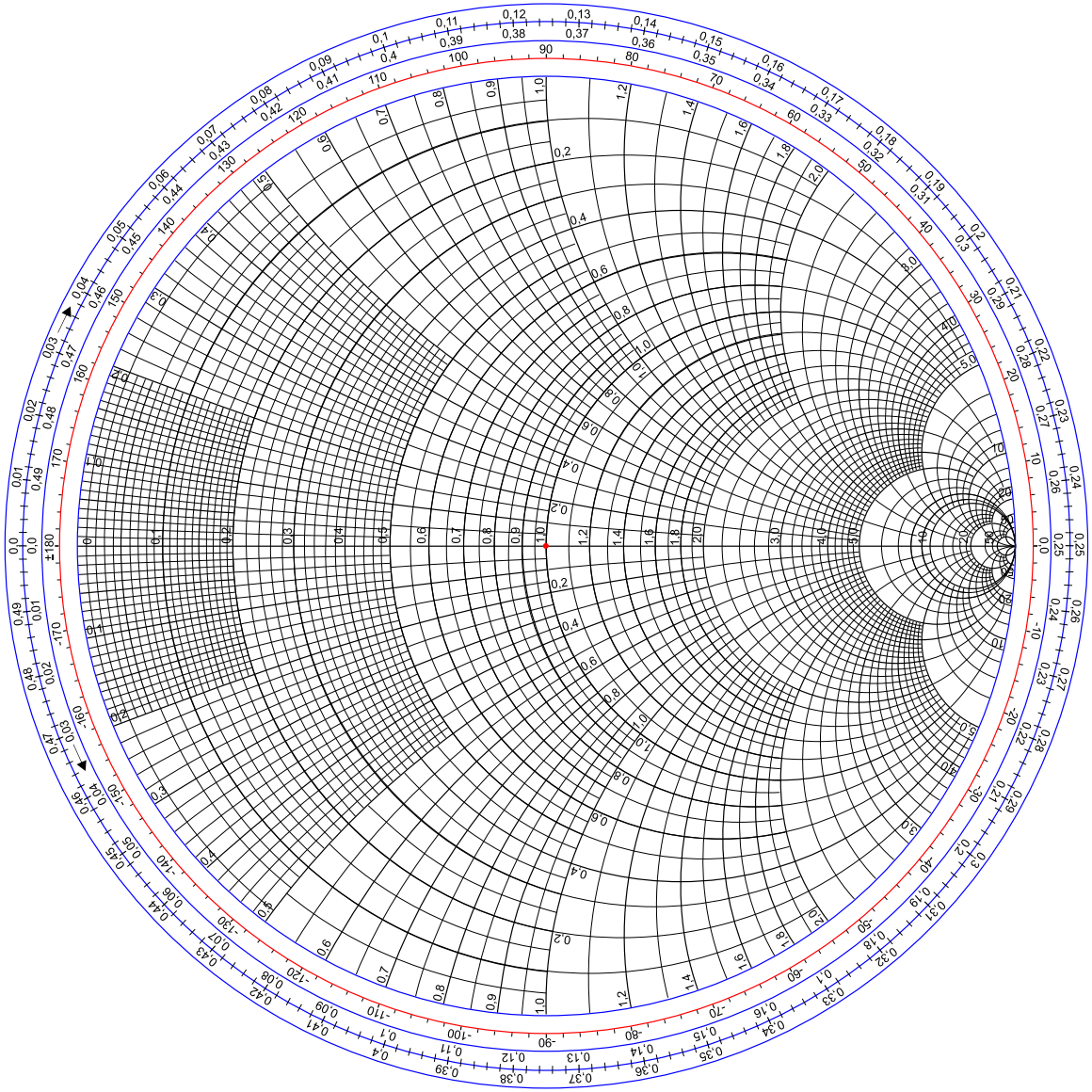
# Impedance Matching Example



## Previous Example

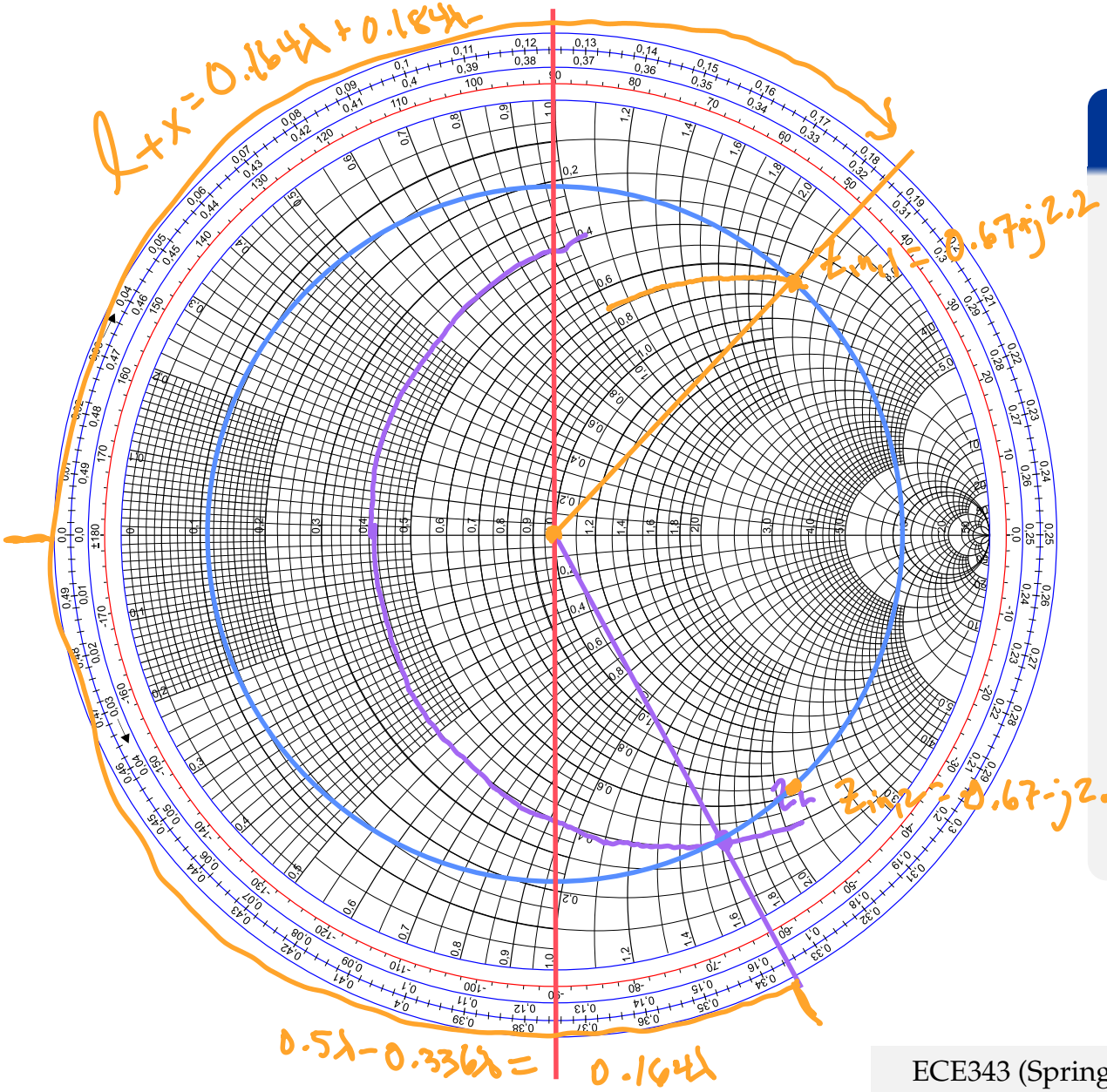
- Match $Z_s$=50Ω to $Z_L$=33.9+j17.9Ω
- $Z_0$ = 50Ω
- $z_L = 0.678 + j0.358$
- _____
- _____
- _____
- _____
- _____

# Impedance Matching Example



# Impedance Matching: Your Turn

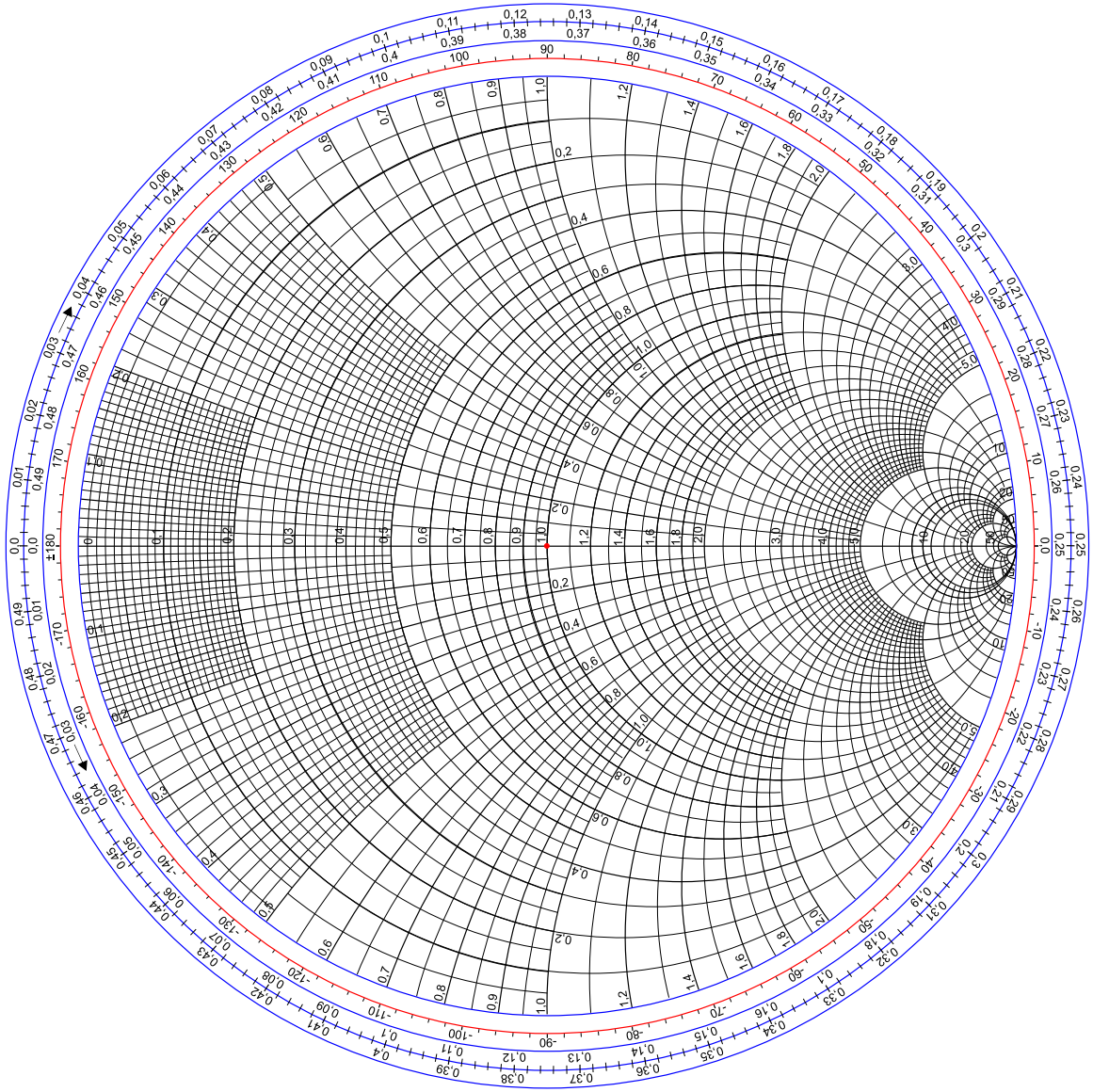
Minimize $l_{TX}$



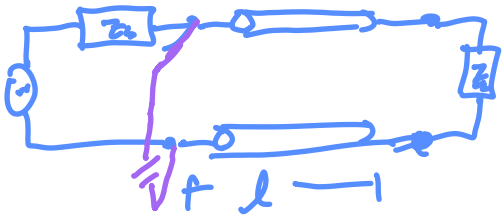

## Example 2

- Match $Z_s$=50Ω to $Z_L$=30-j120Ω
- $Z_0$ = 75Ω , $z_L$ = 0.4-j1.6
- $z_{in,1}$ = 0.67+j2.2
- $l_{TX}$ = 0.348λ
- _____
- _____
- _____

# Impedance Matching: Your Turn



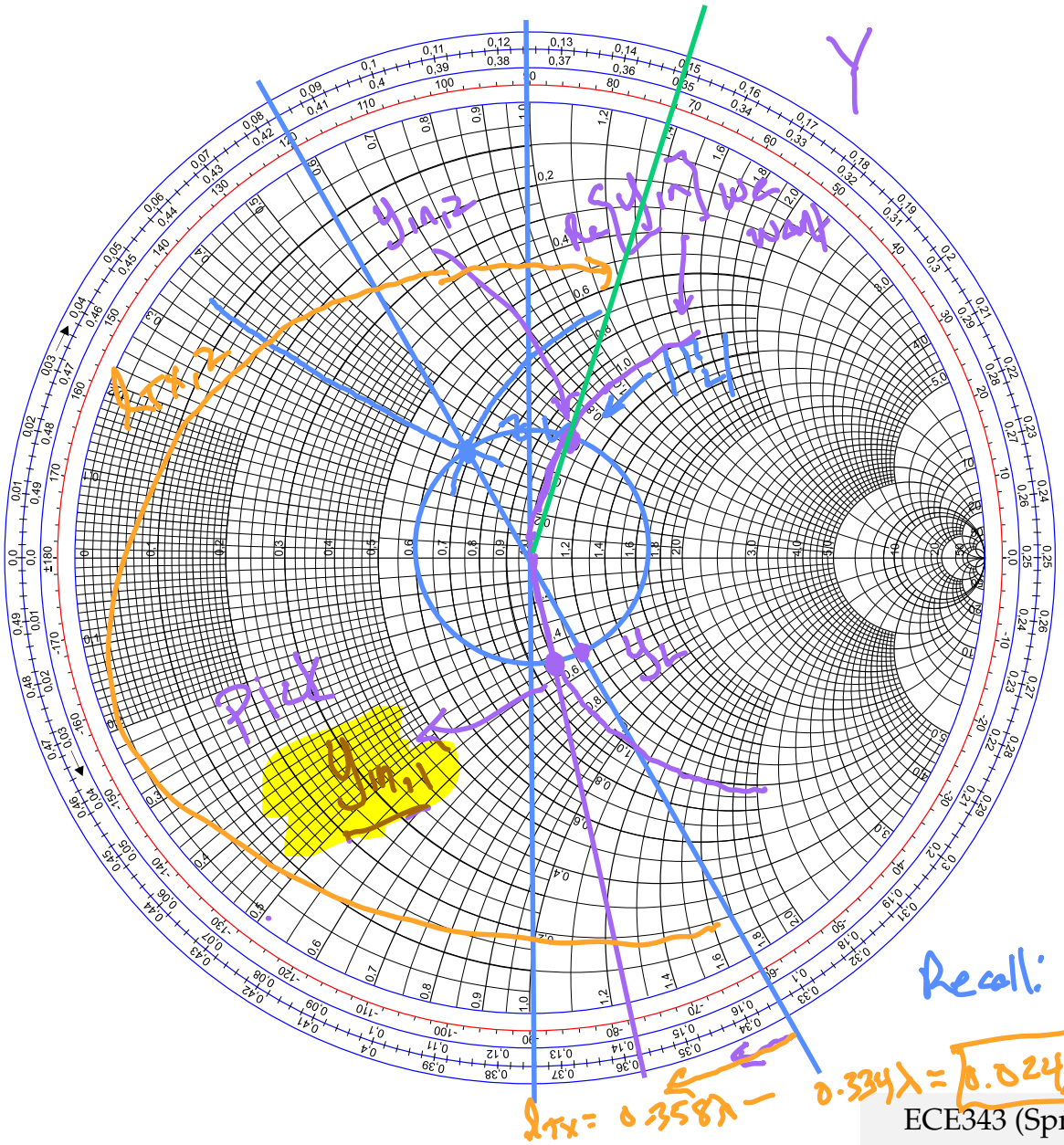
# Parallel Component




$Z \rightarrow Y$

$Y = \frac{1}{Z}$ or ... 180° rotation of S.C.

# Impedance Matching: Parallel Component




## Previous Example

- Use a parallel component
- $Z_L = 0.678 - j0.358$
- Rotate 180° → $y_L = 1.15 + j0.61$
- Verify: $y_L = \frac{1}{0.678 - j0.358} = 1.153 + j0.61$ ✓
- $\mathrm{Re}[y_{in}] = \frac{1}{\mathrm{Re}[z_{in}]} = \frac{1}{1} = 1$ ←
- $y_{in,1} = 1 - j0.525$
- Need $y_c = +j0.525$ ↑ capacitor

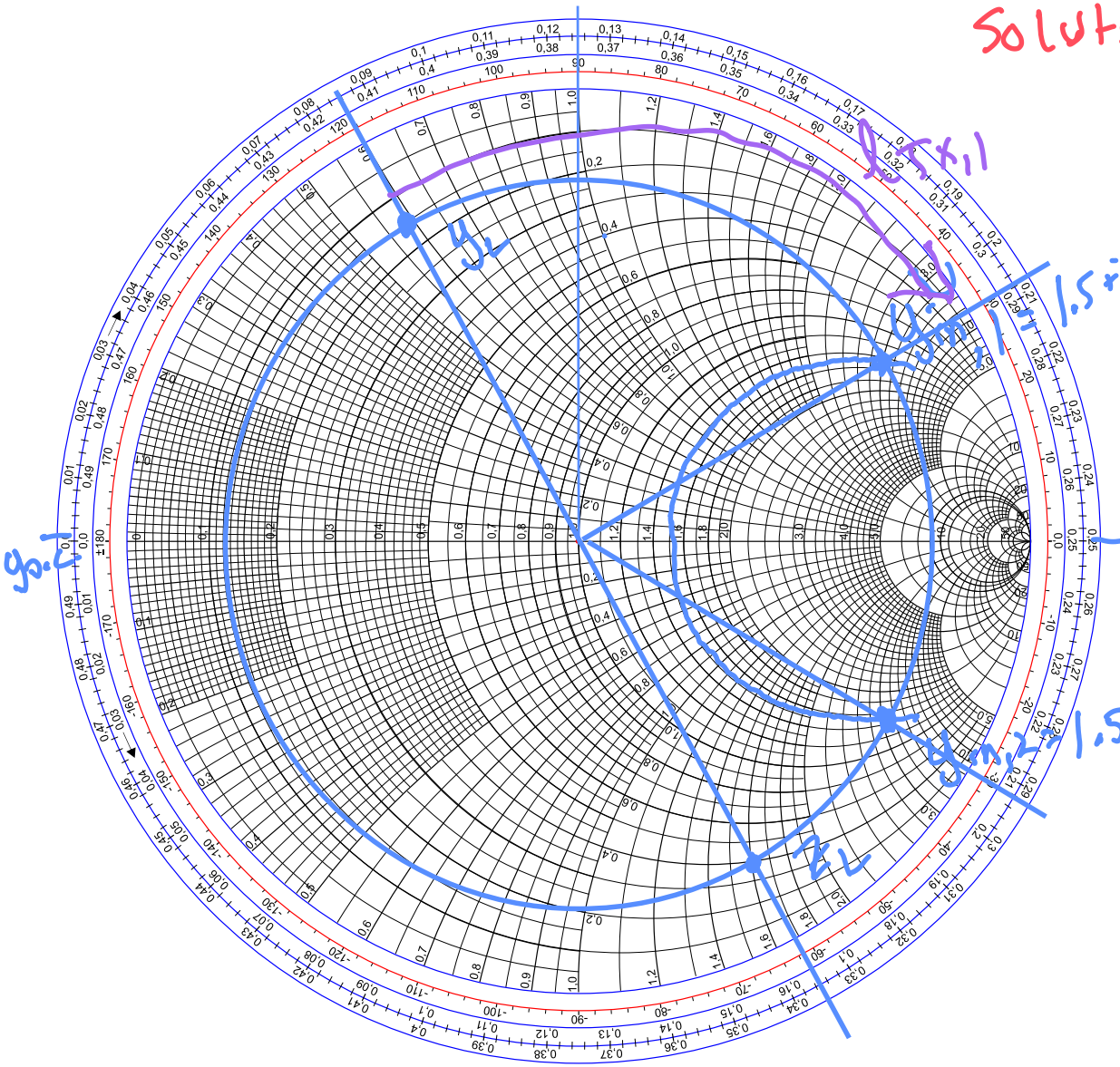
Recall: $y_{ind.} = \frac{1}{j\omega L}$ $y_{cap.} = j\omega C$

# Impedance Matching: [redacted] - Your Turn

Parallel Component

Solution...

S.C.
O.C




## Example 2

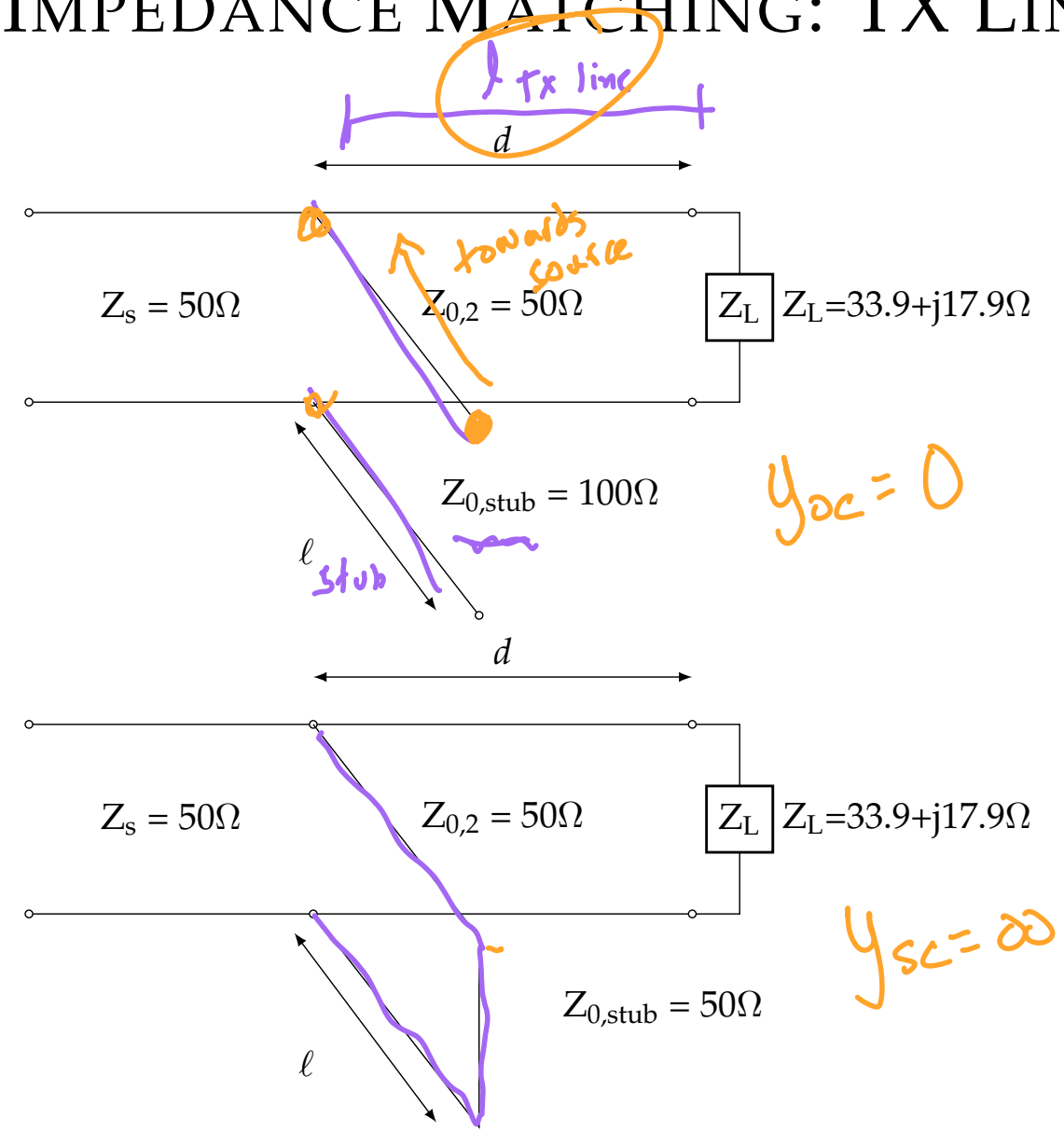
- Match $Z_s=50\Omega$ to $Z_L=$ ~~$33.9+j17.9\Omega$~~ $30-j120\ \Omega$
- _____ $Z_0=75$
- $z_L=0.4-j1.6$
- _____
- $y_L=0.15+j0.58$
- _____
- $y_{in}=\frac{1}{Re[z_s]}=\frac{1}{50/75}=1.5$
- _____
- $y_{in}=1.5\pm j3$
- _____
- $y_{in,1}\rightarrow l_{TX,1}=0.208\lambda$ $-0.058\lambda$
- _____

Need $y_c=-j3$ → inductor $\frac{1}{j\omega L}=-j3*Z_0$

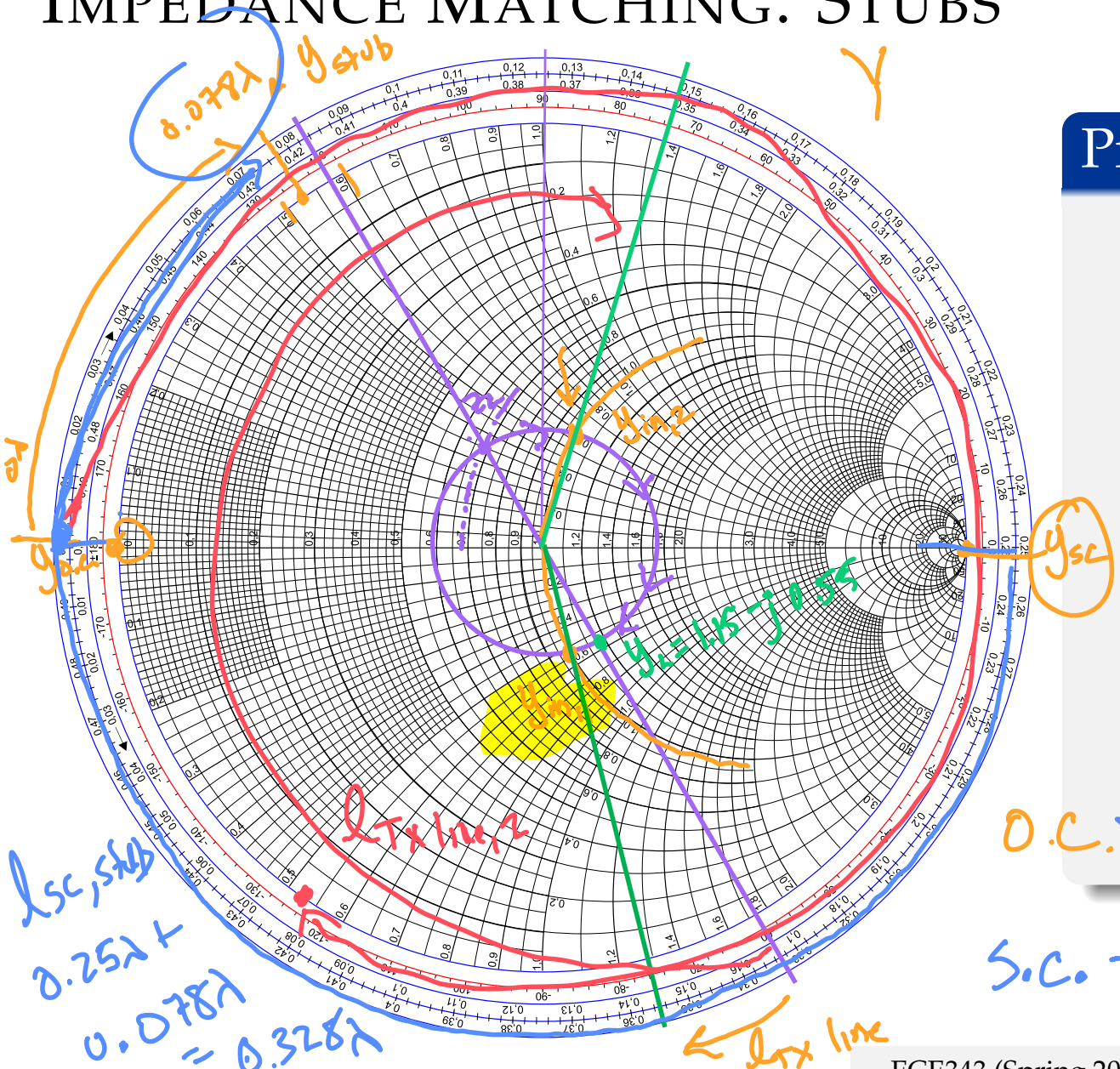
$0.122\lambda$

# Impedance Matching: TX Line Stubs

Instead of lumped



# Impedance Matching: Stubs




## Previous Example

- Match $Z_s = 50\Omega$ to $Z_L = 33.9 + j17.9\Omega$
- $z_L = 0.678 + j0.358$
- $y_L = 1.15 - j0.55$
- $Re[y_m] = \frac{1}{Re[z_L]} = \frac{1}{1} = 1$
- $y_m = 1 \pm j0.525$
- Need $y_{stub} \rightarrow +j0.525$
- O.C. → $l_{stub} = 0.078\lambda$
- S.C. → $l_{stub} = 0.328\lambda$

# Impedance Matching: Parallel Component

stub

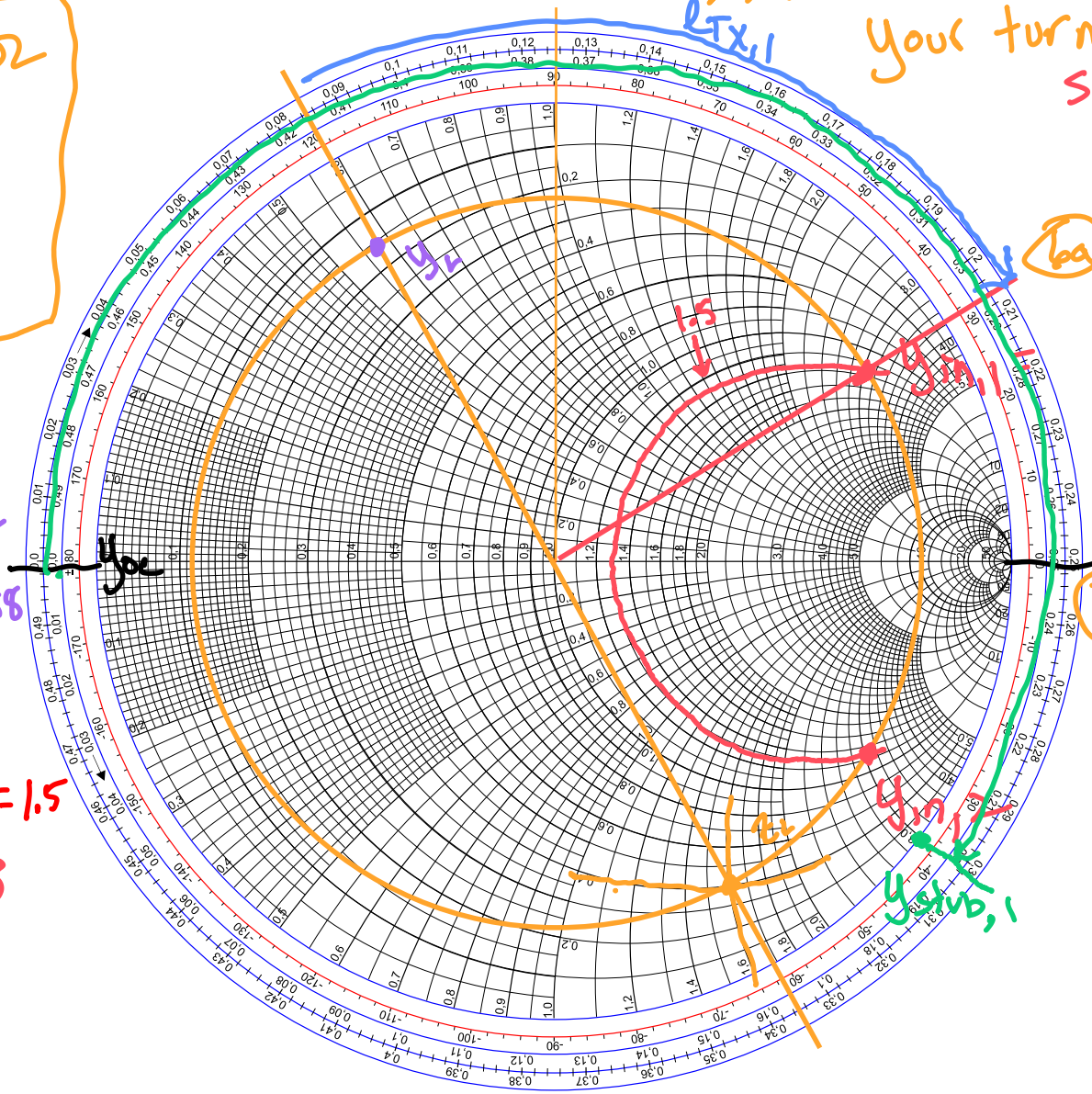
your turn!

solution

$Z_L = 30 - j120\,\Omega$

$Z_S = 50\,\Omega$

$Z_0 = 75\,\Omega$




0 $Z_L = 0.4 - j1.6$

2 $y_L = 0.16 + j0.575$

calc → $0.147 + j0.588$

3 $z_S = \frac{50}{75} = 0.67$

4 $\mathrm{Re}[y_{in}] = \frac{1}{z_S} = 1.5$

5 $y_{in} = 1.5 \pm j3$

6a pick $y_{in,1}$ ...

$l_{TX,1} = 0.206\lambda - 0.085\lambda$

$0.121\lambda$

7a $l_{stub}$ ...

$y_{stub,1} = \mathrm{Im}[y_m^*] = -j3$

O.C. → $0.310\lambda$

S.C → $0.060\lambda$